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Database Design for

Smart Complaint Management System

SMART COMPLAINT MANAGEMENT SYSTEM

A Smart Complaint Management System (SCMS) is a web-based platform designed to bridge the communication gap between citizens and civic authorities. It enables residents to report real-world issues such as road damage, water leakage, garbage problems, and public infrastructure failures through a user-friendly web application.

The system provides authorities with a centralized dashboard to receive, assign, track, and resolve complaints efficiently. Key features include complaint submission with media attachments, GPS location tagging, automated notifications, status tracking, admin dashboards, and citizen feedback after resolution.

Designing a robust database for this system requires capturing entities such as users, wards, categories, complaints, assignments, status updates, media files, feedback, and notifications — along with their relationships.

Entities

The database consists of the following nine core entities:

- users
- wards
- categories
- complaints
- complaint_media
- assignments
- status_updates
- feedback
- notifications

Entity Relationships

- A user (citizen) can submit many complaints, but each complaint belongs to exactly one user — One-to-Many Relationship (users → complaints).
- A category can contain many complaints, but each complaint belongs to one category — One-to-Many Relationship (categories → complaints).
- A ward can have many complaints filed, but each complaint is linked to one ward — One-to-Many Relationship (wards → complaints).
- A complaint can have multiple media attachments (photos/videos), but each media record belongs to one complaint — One-to-Many Relationship (complaints → complaint_media).
- A complaint can be assigned to one authority at a time, and an authority can handle many complaints — One-to-Many Relationship (users → assignments).
- A complaint can have many status updates over its lifecycle, but each update belongs to one complaint — One-to-Many Relationship (complaints → status_updates).
- A resolved complaint can have exactly one feedback entry from the citizen — One-to-One Relationship (complaints → feedback).
- A user can receive many notifications, but each notification is for one specific user — One-to-Many Relationship (users → notifications).
- A ward belongs to one city/state, and a user (citizen) is associated with one ward — referenced via ward_id foreign key.

Entity Attributes

Attributes are the fields of entities that define the characteristics or properties of each entity within the database. Below are the detailed attributes for all entities.

1. users

Stores all registered users including citizens who file complaints and admin/authority users who manage them.

Field Name	Data Type	Constraints	Description
user_id	INT	PK, NOT NULL, AUTO_INCREMENT	Unique identifier for each user
full_name	VARCHAR(100)	NOT NULL	Full name of the user
email	VARCHAR(150)	UNIQUE, NOT NULL	Email address used for login and notifications
password_hash	VARCHAR(255)	NOT NULL	Hashed password for authentication
phone_number	VARCHAR(15)	NULL	Optional contact phone number

Field Name	Data Type	Constraints	Description
role	ENUM	NOT NULL, DEFAULT 'citizen'	Role: citizen, admin, or authority
ward_id	INT	FK → wards(ward_id)	Ward/area the citizen belongs to
is_active	BOOLEAN	DEFAULT TRUE	Whether the account is active
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Account creation timestamp
updated_at	DATETIME	ON UPDATE CURRENT_TIMESTAMP	Last profile update timestamp

2. wards

Represents geographical areas/wards for which complaints are raised and authorities are assigned.

Field Name	Data Type	Constraints	Description
ward_id	INT	PK, NOT NULL, AUTO_INCREMENT	Unique identifier for a ward
ward_name	VARCHAR(100)	NOT NULL	Name of the ward or locality
city	VARCHAR(100)	NOT NULL	City to which the ward belongs
state	VARCHAR(100)	NOT NULL	State to which the ward belongs
pincode	VARCHAR(10)	NOT NULL	PIN code of the ward area

3. categories

Defines the types of complaints that can be filed (e.g., Road Damage, Water Leakage, Garbage).

Field Name	Data Type	Constraints	Description
category_id	INT	PK, NOT NULL, AUTO_INCREMENT	Unique identifier for the complaint category
category_name	VARCHAR(100)	UNIQUE, NOT NULL	Name of the complaint category
description	TEXT	NULL	Brief description of the complaint type
icon_url	VARCHAR(255)	NULL	URL to category icon for UI display
is_active	BOOLEAN	DEFAULT TRUE	Whether this category is available for selection

4. complaints

Core entity that stores every complaint submitted by citizens. Links to user, category, and ward.

Field Name	Data Type	Constraints	Description
complaint_id	INT	PK, NOT NULL, AUTO_INCREMENT	Unique identifier for the complaint
title	VARCHAR(200)	NOT NULL	Short title of the complaint
description	TEXT	NOT NULL	Detailed description of the issue reported
user_id	INT	FK → users(user_id)	ID of the citizen who filed the complaint
category_id	INT	FK → categories(category_id)	Type/category of the complaint
ward_id	INT	FK → wards(ward_id)	Ward/area where the issue was reported
address	VARCHAR(300)	NULL	Exact address or landmark of the issue
latitude	DECIMAL(10,8)	NULL	GPS latitude for location pinning on map
longitude	DECIMAL(11,8)	NULL	GPS longitude for location pinning on map
priority	ENUM	DEFAULT 'medium'	Priority level: low, medium, high, critical
status	ENUM	DEFAULT 'pending'	Status: pending, assigned, in_progress, resolved, rejected, closed
submitted_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Date and time the complaint was submitted
resolved_at	DATETIME	NULL	Date and time the complaint was resolved
updated_at	DATETIME	ON UPDATE CURRENT_TIMESTAMP	Last update timestamp

5. complaint_media

Stores photo or video attachments uploaded by citizens along with a complaint.

Field Name	Data Type	Constraints	Description
media_id	INT	PK, NOT NULL, AUTO_INCREMENT	Unique identifier for each media file
complaint_id	INT	FK → complaints(complaint_id)	Complaint the media file is attached to
file_url	VARCHAR(500)	NOT NULL	URL or file path to the stored media
file_type	ENUM	NOT NULL	Type of file: image or video
uploaded_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Upload timestamp

6. assignments

Tracks which authority/official is assigned to handle a specific complaint.

Field Name	Data Type	Constraints	Description
assignment_id	INT	PK, NOT NULL, AUTO_INCREMENT	Unique identifier for the assignment record
complaint_id	INT	FK → complaints(complaint_id)	The complaint being assigned
assigned_to	INT	FK → users(user_id)	Authority user responsible for resolution
assigned_by	INT	FK → users(user_id)	Admin who performed the assignment
assigned_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Timestamp of the assignment action
notes	TEXT	NULL	Admin notes or instructions for the authority

7. status_updates

Maintains a complete audit trail of every status change made on a complaint.

Field Name	Data Type	Constraints	Description
update_id	INT	PK, NOT NULL, AUTO_INCREMENT	Unique identifier for each status update record
complaint_id	INT	FK → complaints(complaint_id)	Complaint whose status was changed
updated_by	INT	FK → users(user_id)	User (admin/authority) who made the update
previous_status	ENUM	NOT NULL	Status before the update
new_status	ENUM	NOT NULL	Status after the update
remarks	TEXT	NULL	Optional remarks explaining the status change
updated_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Timestamp of the status change

8. feedback

Captures citizen satisfaction ratings and comments after a complaint is resolved.

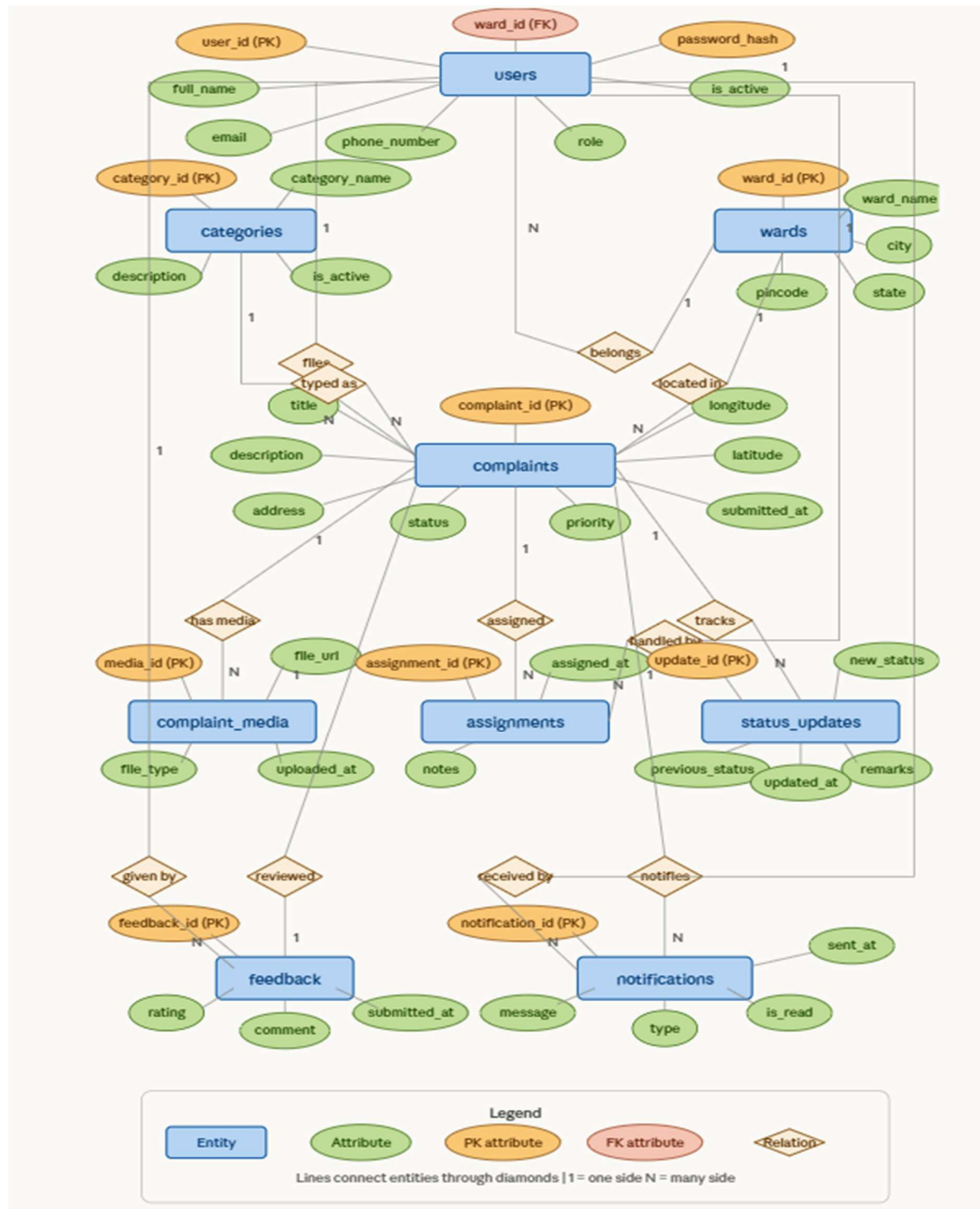
Field Name	Data Type	Constraints	Description
feedback_id	INT	PK, NOT NULL, AUTO_INCREMENT	Unique identifier for the feedback record
complaint_id	INT	FK → complaints(complaint_id), UNIQUE	Resolved complaint being rated (one feedback per complaint)
user_id	INT	FK → users(user_id)	Citizen who submitted the feedback
rating	TINYINT	CHECK (rating BETWEEN 1 AND 5)	Satisfaction rating from 1 (poor) to 5 (excellent)
comment	TEXT	NULL	Optional written feedback or suggestion
submitted_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Timestamp when feedback was submitted

9. notifications

Manages system-generated notifications sent to users regarding complaint activity.

Field Name	Data Type	Constraints	Description
notification_id	INT	PK, NOT NULL, AUTO_INCREMENT	Unique identifier for each notification
user_id	INT	FK → users(user_id)	User who receives the notification
complaint_id	INT	FK → complaints(complaint_id), NULL	Related complaint (nullable for system alerts)
message	TEXT	NOT NULL	Notification message content
type	ENUM	NOT NULL	Type: status_update, assignment, feedback_request, alert
is_read	BOOLEAN	DEFAULT FALSE	Whether the notification has been read
sent_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Timestamp when the notification was sent

ENTITY RELATIONSHIP DIAGRAM — SMART COMPLAINT MANAGEMENT SYSTEM



ENHANCED ENTITY RELATIONSHIP DIAGRAM

